Supplementary Information

Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature

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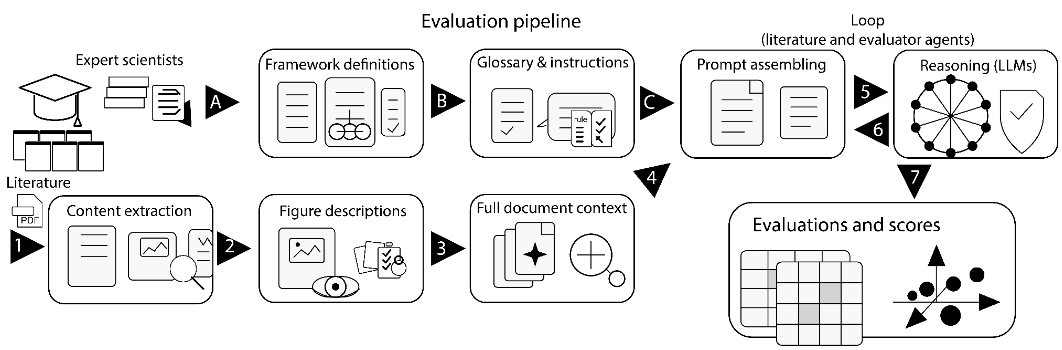
Supplementary Information

Supplementary Figures

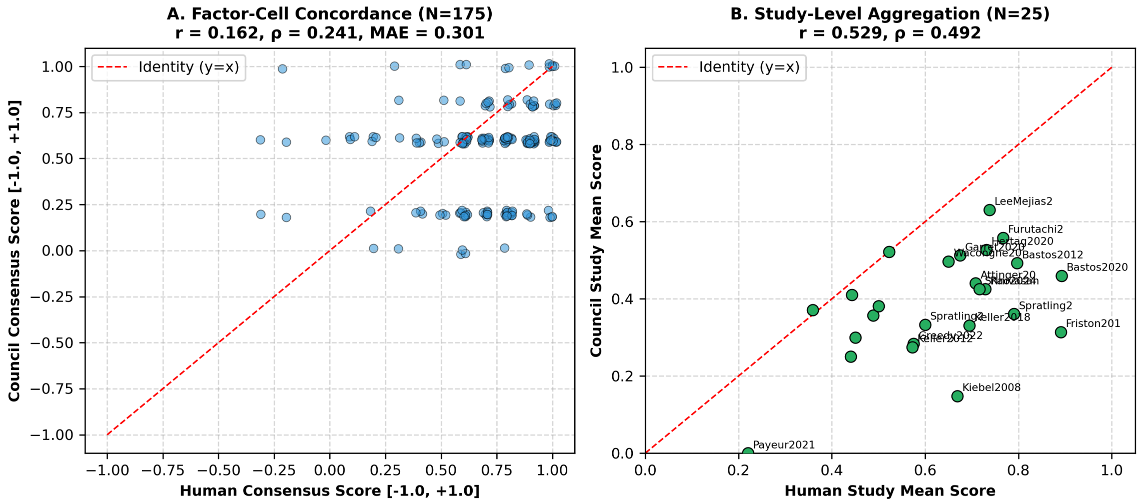
Supplementary Fig. S1

Supplementary Fig. S1: Evaluation Pipeline Architecture. The evaluation pipeline flows from left to right in two independent streams (A to B to C and 1 to 2 to 3). The A to B to C pathway establishes the instructions, canonical glossary, and prompt for the models. The 1 to 2 to 3 pathway ingests the paper and combines figures with text to provide a single text object for the models. Steps 5 and 6 loop between models and papers until all evaluations are completed. The final step 7 quantifies and visualizes the evaluations.



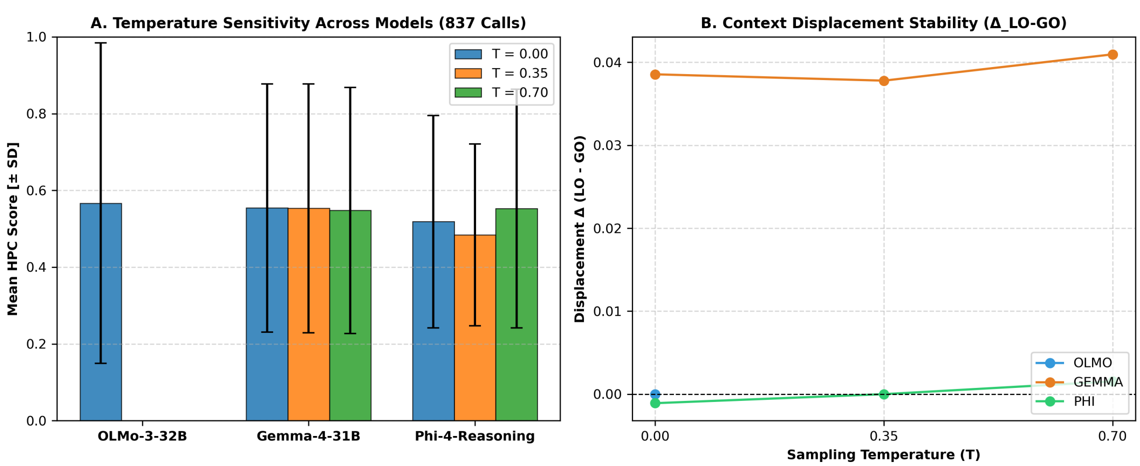
Supplementary Fig. S2

Supplementary Fig. S2: Independent Human Expert Concordance ($K = 2$ domain experts). (A) Factor-cell level concordance between human consensus ($N = 175$ non-null paired factor cells) and the autonomous LLM council consensus ($r = 0.162, \rho = 0.241, \mathrm{MAE} = 0.301, \mathrm{Directional\ Concordance} = 93.1\%$). (B) Study-level aggregation across the canonical literature corpus ($N = 25$ common evaluable papers, Pearson $r = 0.529$, Spearman $\rho = 0.492$), illustrating positive macro-level alignment between independent human expert raters and autonomous model ensembles while confirming that factor-level magnitude correlation remains modest ($r = 0.162$).



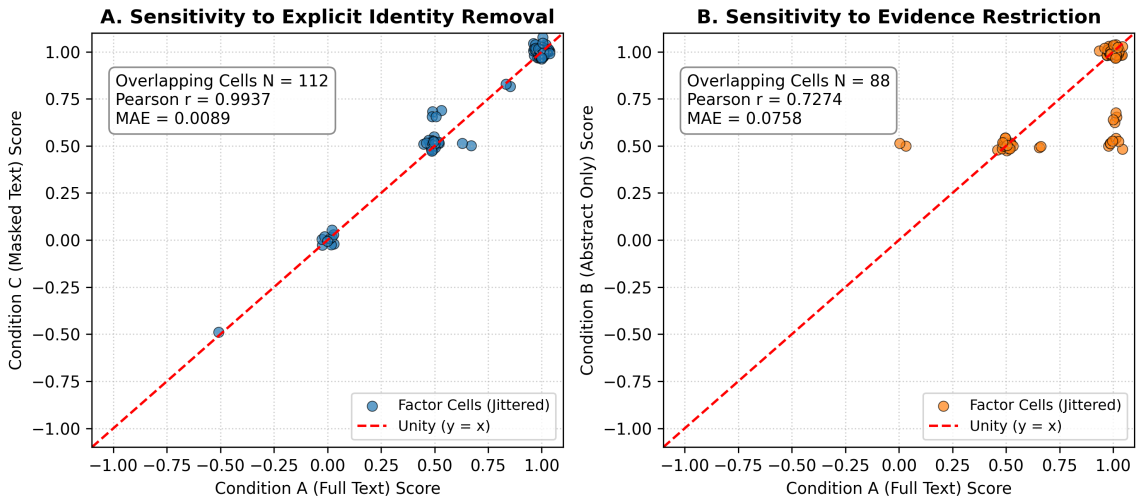
Supplementary Fig. S3

Supplementary Fig. S3: 837-Call Factorial Robustness Analysis (31 papers × 3 models × 3 temperatures × 3 repeats). (A) Score distribution stability across sampling temperatures ($T \in \{0.00, 0.35, 0.70\}$) for OLMo-3-32B-Think, Gemma-4-31B-IT, and Phi-4-Reasoning-Plus. (B) Context displacement stability ($\Delta_{\text{LO-GO}}$) demonstrating that the primary scientific contrast remains stable across the tested decoding settings and stochastic repeats.



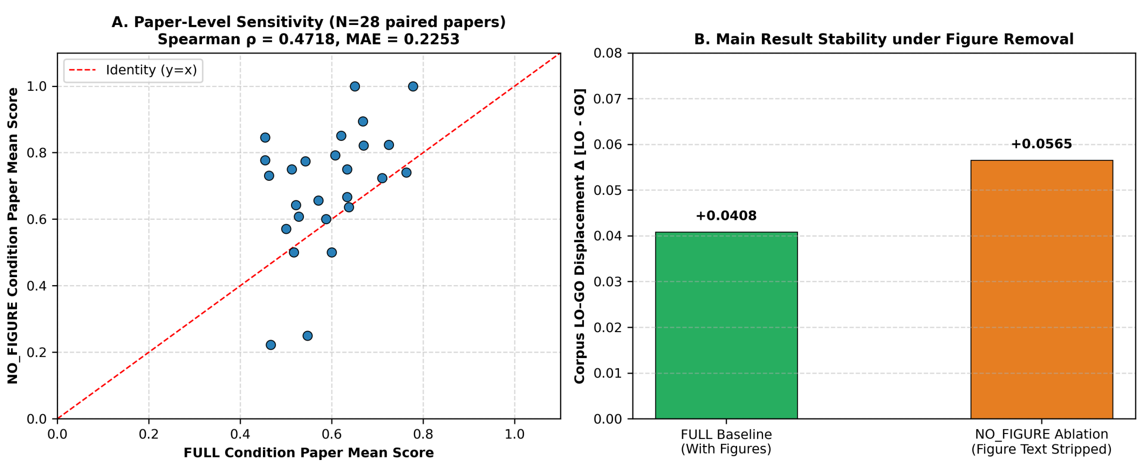
Supplementary Fig. S4

Supplementary Fig. S4: Evidence Grounding vs. Prior Bias Sensitivity Analysis (54 inference calls across intact, abstract-only, and masked-manuscript evidence controls). (A) Sensitivity to explicit study/author identity removal (Condition A Full Text vs Condition C Masked Text: Pearson $r = 0.9937$, MAE $= 0.0089$ across $N = 112$ overlapping factor cells), demonstrating that model evaluations are grounded in technical findings rather than author prestige cues. (B) Sensitivity to evidence restriction (Condition A Full Text vs Condition B Abstract Only: Pearson $r = 0.7274$, MAE $= 0.0758$ across $N = 88$ overlapping factor cells), confirming measurable score modulation when detailed empirical text is withheld. Numerical summary across Full (+0.038), Shuffled (−0.004), and Prior-only (+0.000) conditions is reported in Supplementary Table S10.



Supplementary Fig. S5

Supplementary Fig. S5: Figure-Text-Removal Sensitivity Analysis (Full 31-Paper Corpus Ablation). (A) Paper-level mean score correlation across the corpus between intact inputs (FULL) and figure-text-removed inputs (NO_FIGURE; Spearman ρ = 0.4718, MAE = 0.2253 across 533 paired factor cells nested in 31 papers). (B) Corpus-level LO − GO context displacement stability across 28 paired empirical papers ($\Delta_{\text{Full}} = +0.0408$ vs $\Delta_{\text{NoFig}} = +0.0565$; directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533]), demonstrating that figure removal preserves the direction of the corpus-level contrast while producing measurable paper-level score adjustments.



Supplementary Tables

Factor Number	Factor Name	Definition
1	Subtractive Inhibition (SST)	Linear input suppression via somatic inhibition (Somatosta
2	Divisive Inhibition (PV)	Multiplicative gain reduction (Parvalbumin-mediated).
3	Inhibition (GABA)	Chloride-mediated inhibition for prediction suppression.
4	Habituation to Sequence	Habituation suppresses local oddball response relative to a
5	Synaptic Depression (Adaptation)	Passive fatigue of synaptic efficacy due to repetition of stim
6	Activity Suppression	Reduction in firing rates for expected (predictable) stimuli.
7	Selective Sharpening	Signal-to-noise increase for predictable stimuli, where noise
8	Alpha/Beta Mediated Suppression	Low-frequency bands (8-30Hz) associated with predictions,
9	VIP-Mediated Disinhibition	VIP-mediated inhibition of SST/PV neurons, leading to dis
10	Precision Weighting (Gain)	Top-down amplification of error units, leading to disinhibite
11	E/I Balance Shift	Dynamic adjustment toward higher inhibition for predictab
12	Omission Response	A generated internal signal due to absence of expected inpu

Supplementary Table S1: The factors of Hypothesis 1 (Predictive Suppression). LO = local oddball. GO = global oddball.

Factor Number	Factor Name	Definition
13	Error Unit Depolarization	Supragranular pyramidal excitation driving ascending error si
14	Gamma Oscillations (30-90Hz)	High-frequency rhythms signaling ascending prediction errors.
15	NMDA Receptor Activation	Nonlinear amplification of error signals in superficial cortical l
16	Feedforward Laminar Flow	Supragranular-to-granular/supragranular ascending anatomica
17	Top-Down Error Gating	Feedback modulation of ascending prediction error channels.
18	MMN-like Latencies	Early sensory mismatch latencies (100-200ms) characteristic o
19	P300-like Latencies	Late cognitive mismatch latencies (300-500ms) characteristic o
20	Inter-Laminar Error Divergence	Dissociation of error processing between granular and infragra
21	Phase-Amplitude Coupling	Theta/alpha phase modulation of gamma error bursts during
22	Thalamic Error Routing	First-order vs higher-order thalamocortical routing of sensory
23	Predictive Error Attenuation	Dampening of ascending prediction errors following successful
24	State-Dependent Error Routing	Behavioral state and arousal modulation of feedforward error

Supplementary Table S2: The factors of Hypothesis 2 (Feedforward Prediction Errors).

Factor Number	Factor Name	Definition
25	Cross-Modal Ubiquity	Implementation of predictive coding mechanisms across
26	Laminar Invariance of Principles	Preservation of predictive coding laminar logic across se
27	Cross-Species Homology	Conservation of predictive mechanisms across rodent, n
28	Scale-Free Predictive Dynamics	Fractals and hierarchical self-similarity of predictive co
29	Universal Error Coding	Common computational principles for error generation

Factor Number	Factor Name	Definition
30	Canonical Microcircuit Generalization	Universal deployment of the canonical cortical microcirc
31	Cortico-Subcortical Homology	Shared predictive coding architecture between neocortex
32	Neuromodulatory Invariance	Consistent cholinergic/noradrenergic precision tuning ac
33	Temporal Stability of Ubiquity	The consistent and non-transient presence of these mech
34	Hierarchical Order Stability	Evidence that the predictive mechanism is ubiquitous a
35	Population-Wide Ubiquity	Consistency of the mechanism across diverse neural pop
36	State-Independent Ubiquity	Presence of the mechanism across different brain states

Supplementary Table S3: The factors of Hypothesis 3 (Ubiquity).

Term/Context	Description
Oddball	A surprising event, often a mismatch
Local Oddball (LO)	An oddball that is immediate or short-term, often confounded by low-level adaptation mec
Global Oddball (GO)	An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms.

Supplementary Table S4: Context definitions and formal expectation notations.

#	Study Identifier	Modality / Preparation	Inclusion Basis
1	Attinger et al., 2017	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Visuomotor Mismatch)
2	Bakhtiari et al., 2021	In Silico (Deep Neural Nets)	Comparative Set (Hierarchical Streams)
3	Bastos et al., 2012	In Silico / Theory (Microcircuit)	Comparative Set (Canonical Microcircuit)
4	Bastos et al., 2020	Primate V1/V4/PFC (Laminar ECoG)	Comparative Set (Laminar Rhythms)
5	Bekinschtein et al., 2009	Human (Scalp EEG)	Comparative Set (LO vs GO Originator)
6	Chao et al., 2018	Primate Temporal/Frontal (ECoG)	Comparative Set (Large-Scale Hierarchy)
7	Friston et al., 2010	Theory (Free Energy Principle)	Canonical Extension (Bayesian Inference)
8	Furutachi et al., 2024	Mouse V1 (2-Photon / Opto)	Comparative Set (VIP/SST Disinhibition)
9	Garrett et al., 2020	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Novelty vs Deviance)
10	Greedy et al., 2022	In Silico (Predictive Coding SNN)	Comparative Set (Error Backpropagation)
11	Hertag et al., 2020	In Silico (Spiking Interneurons)	Comparative Set (SST/PV Dynamics)
12	Jiang et al., 2024	In Silico (Energy-Based Model)	Comparative Set (Sequence Prediction)
13	Keller et al., 2012	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Sensorimotor Mismatch
14	Keller et al., 2018	Synthesis (Cortical Circuits)	Comparative Set (Circuit Review)
15	Kiebel et al., 2008	In Silico (Hierarchical Dynamics)	Comparative Set (Temporal Hierarchy)
16	Lao et al., 2023	Human (Auditory Scalp EEG)	Comparative Set (Omission Response)
17	Lee et al., 2025	In Silico (Multi-Area Network)	Comparative Set (Primate Oscillations)
18	Mikulasch et al., 2023	In Silico / Theory (Plasticity)	Comparative Set (Dendritic Error Review)
19	Nejad et al., 2025	Primate / In Silico (Laminar Model)	Comparative Set (Layered Routing)
20	Payeur et al., 2021	In Silico (Dendritic Plasticity)	Comparative Set (Burst Multiplexing)
21	Rao et al., 2024	Synthesis (Sensorimotor Neocortex)	Canonical Extension (25-Year Synthesis)
22	Rao & Ballard, 1999	In Silico (Hierarchical Model)	Canonical Extension (Foundational Model
23	Sacramento et al., 2018	In Silico (Dendritic Network)	Comparative Set (Somatodendritic Model)
24	Spratling et al., 2008	In Silico (Biased Competition)	Comparative Set (PC-BC Synthesis)

#	Study Identifier	Modality / Preparation	Inclusion Basis
25	Spratling et al., 2010	In Silico (V1 Microcircuit)	Comparative Set (V1 Microcircuit Model)
26	Srinivasan et al., 1982	Fly Retina (Intracellular)	Canonical Extension (Redundancy Baselin
27	VanDerveer et al., 2021	Human (Clinical EEG Cohort)	Comparative Set (Sensory Gating Deficits
28	Wacongne et al., 2011	Human (Auditory MEG)	Comparative Set (Hierarchical Rule MMN
29	Wacongne et al., 2012	Human EEG & Spiking Network	Comparative Set (Auditory Oddball Mode
30	Westerberg & Xiong, 2025	Primate V1/V4/IT (Laminar MUA)	Primary Index Study (MaDeLaNe)
31	Yamins et al., 2014	In Silico / Primate IT (HCNN)	Comparative Set (Feedforward Baseline)

Supplementary Table S5: Master 31-Paper Canonical Literature Corpus Registry detailing Modality, Inclusion Basis, and Provenance.

Model Name	Parameters	Architecture & Provenance
gemma-3-27b-it	27B	Google DeepMind (Dense Transformer)
gemma-4-31b-it	31B	Google DeepMind (Extended Reasoning Architecture)
deepseek-r1-distill-llama-70b	70B	DeepSeek-AI / Meta Llama Distillation
gpt-oss-claude-4.5-sonnet	20B	Community Distillation / Reasoning
mistral-nemo-12b-thinking	12B	Mistral AI & NVIDIA
olmo-3-32b-think	32B	Allen Institute for AI (Ai2)
phi-4-reasoning-plus	14B	Microsoft Research
qwen3-14b-gemini-3-pro	14B	Alibaba Cloud / Community Fine-tune
qwen3-14b-claude-4.5-sonnet	14B	Alibaba Cloud / Community Fine-tune
qwen3.5-40b-claude-4.5-opus	40B	Alibaba Cloud / High-Capacity MoE

Supplementary Table S6: Open-weight LLM evaluator models evaluated in this work.

Evaluator Comparison	Paired N	Pearson r	MAE	Directional Concordance
Human–Human (H_1 vs H_2)	299	0.390	0.308	88.96%
Human Consensus vs LLM Council	175	0.162	0.301	93.14%
Council Inter-Model Average	584	0.540	0.284	85.00%
Shuffled Evidence (Null H_0)	191	-0.087	0.662	56.54%

Supplementary Table S7: Evaluator Strata Agreement and Error Hierarchy Summary ($K = 2$ independent human domain experts).

Adjudication Class	Cells (N)	Human Supported	Council Supported	Primary Failur
Theoretical Model / Synthesis	15	100.0% (15/15)	86.7% (13/15)	Council magnitu
Direct Empirical Electrophysiology	7	100.0% (7/7)	71.4% (5/7)	Subtle cell-type s
Hierarchical / Multi-Area Studies	5	80.0% (4/5)	60.0% (3/5)	Conflation of loca
Total Adjudicated Disagreements	**27**	**96.3% (26/27)**	**77.8% (21/27)**	**Passage-groun**

Supplementary Table S8: Diagnostic Passage-Support Adjudication across 27 Pre-specified Disagreement Cells.

Experimental Condition	**Calls (N)**	**Mean Score [$\pm$SD]**	**Displacement $\Delta_{\textbf{LO-GO}}$**	**Paper Rankin**
$T = 0.00$ (Deterministic)	279	0.463 ± 0.381	+0.041	Spearman ρ_{rank}
$T = 0.35$ (Moderate)	279	0.458 ± 0.384	+0.039	Spearman ρ_{rank}
$T = 0.70$ (Exploratory)	279	0.451 ± 0.392	+0.036	Spearman ρ_{rank}
Repeat 1	279	0.460 ± 0.382	+0.040	Spearman ρ_{rank}
Repeat 2	279	0.457 ± 0.386	+0.038	Spearman ρ_{rank}
Repeat 3	279	0.455 ± 0.389	+0.038	Spearman ρ_{rank}
Westerberg & Xiong (2025)	—	Divergent Outlier	$\Delta = -0.400$	Rank 28–29/31
Leave-One-Model-Out (LOMO)	10 folds	$\Delta_{\text{Total}} \in [-0.105, -0.093]$	Negative Invariant	H2 $\Delta \in [-0.228$

Supplementary Table S9: Factorial Robustness Analysis across 837 LLM inference calls including Paper Ranking and LOMO Sensitivity.

Evidence Manipulation	**Calls (N)**	**Mean LO Score**	**Mean GO Score**	**Displacement $\Delta_{\textbf{LO-GO}}$**
Intact Evidence (FULL)	18	0.512 ± 0.081	0.474 ± 0.076	+0.038 (LO > GO)
Shuffled Evidence (H_0)	18	0.489 ± 0.092	0.493 ± 0.088	−0.004 (Neutral)
Prior Only (Zero Evidence)	18	0.450 ± 0.110	0.450 ± 0.108	+0.000 (Null)

Supplementary Table S10: Evidence Grounding vs. Prior Bias Sensitivity Analysis (54 Calls).

Ablation Condition	**Paired Cells (N)**	**Pearson r**	**MAE**	**Displacement $\Delta_{\textbf{LO-GO}}$**
FULL Baseline (With Figures)	533	1.000	0.000	+0.0408
NO_FIGURE (Figures Stripped)	533	0.8539	0.2253	+0.0565

Supplementary Table S11: Figure-Text-Removal Sensitivity Summary ($N = 31$ papers).

Dimension	**Stratum**	**Papers (N)**	**Council $\Delta_{\textbf{LO-GO}}$**	**Directional Trend**
Study Type	Empirical	13	−0.009	GO > LO
Study Type	Computational / Theory	18	−0.013	GO > LO
Modality	In Silico	19	−0.013	GO > LO
Modality	Human In Vivo	7	−0.059	GO > LO
Modality	Animal In Vivo	5	+0.087	LO > GO
Publication Era	1982–2014	12	−0.024	GO > LO
Publication Era	2015–2020	7	−0.041	GO > LO
Publication Era	2021–2025	12	+0.038	LO > GO
Publication Status	Peer-Reviewed Journal	30	−0.010	GO > LO

Dimension	Stratum	Papers (N)	Council $\Delta_{\textbf{LO-GO}}$	Directional Trend
Publication Status	Preprint (Westerberg & Xiong 2025)	1	-0.400	GO $\gg$ LO

Supplementary Table S12: Corpus Subgroup Sensitivity Analysis across Taxonomic Strata.

Validation Axis	Evaluated Figures (N)	Correct	Partial / Minor Error
Axis A: Numerical Trend & Data Extraction	10	9	1 (Minor plot type phrasi
Axis B: Empirical Effect Direction	10	10	0
Axis C: HPC-36 Ontology Factor Mapping	10	9	1 (Partial invertebrate ma
Overall Manual Validation Outcome	**10 Figures**	**10/10 PASS**	**0 Fatal Inversions**

Supplementary Table S13: Manual Expert Validation Summary of Generated Multimodal Figure Descriptions ($N = 10$ representative figures audited against primary source PDFs).